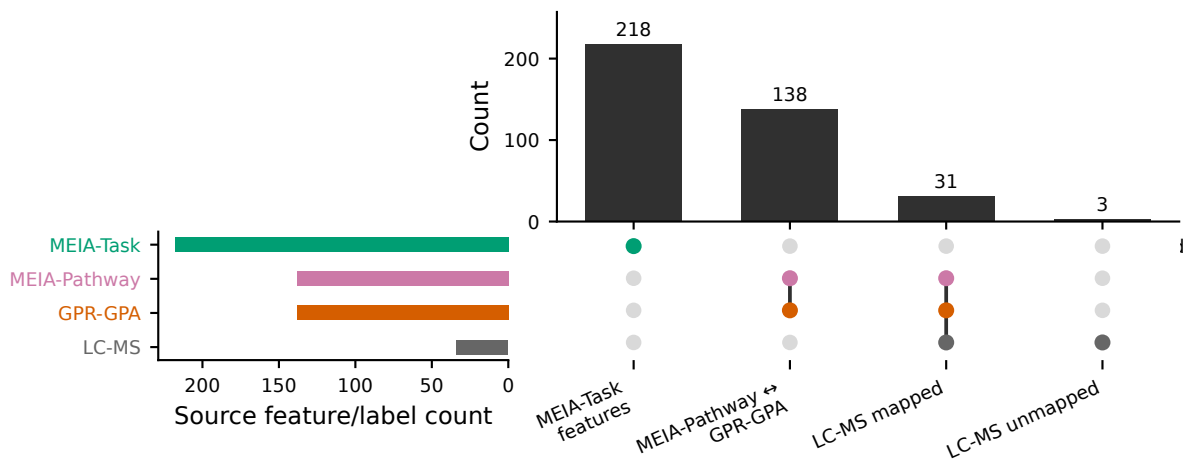
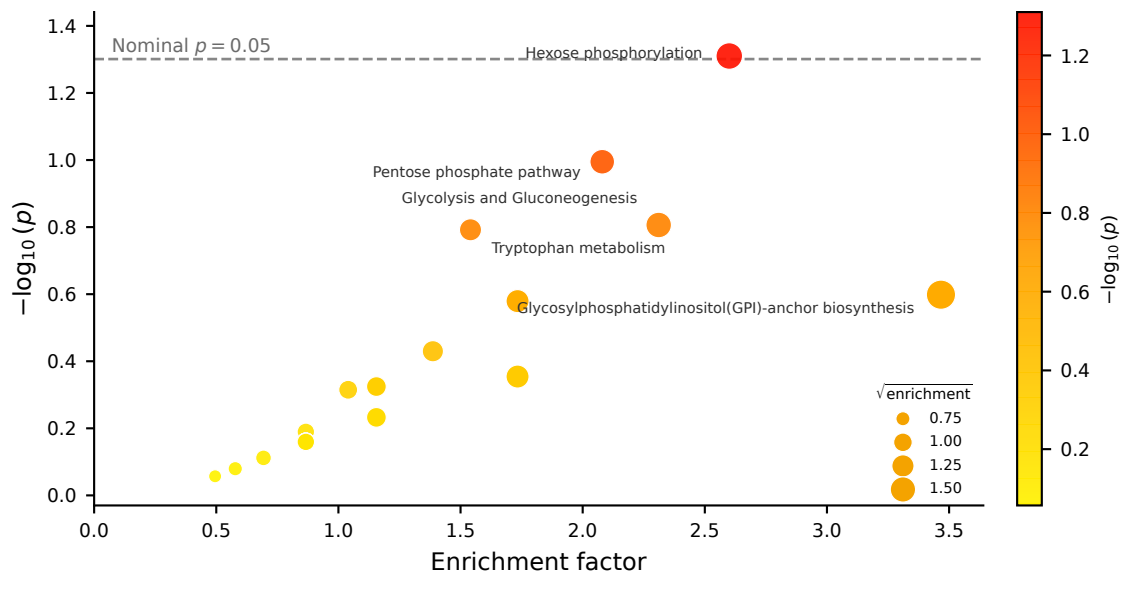
SA Metastatic-status four-set metabolic overlap

Blank-corrected LC-MS pathways: 31/34 mapped; 3 unmapped. MEIA = 356 features (218 tasks + 138 pathways).



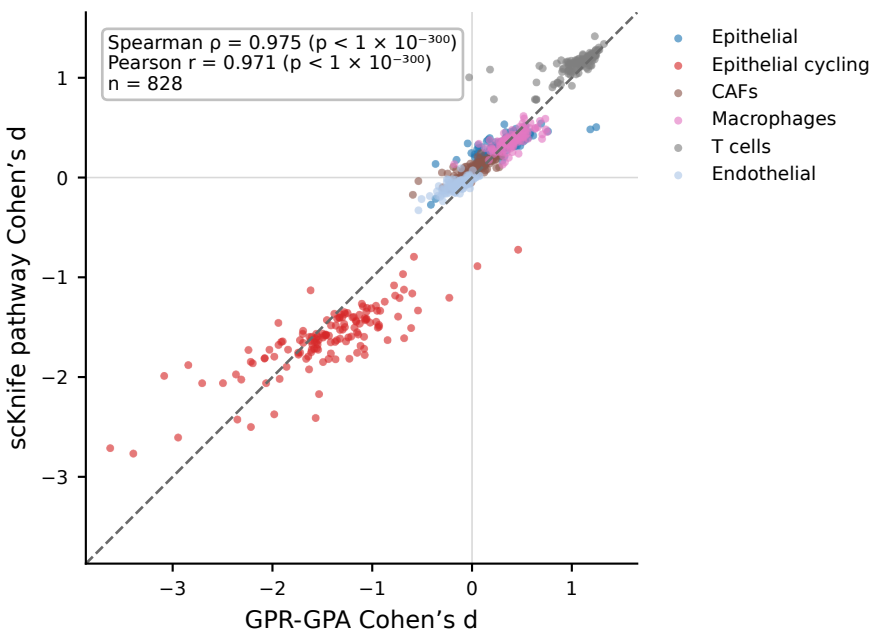
SB Metastatic-status pathway enrichment

34 integrated Mummichog candidates; top five labeled; 0 pass pathway BH-FDR < 0.05.



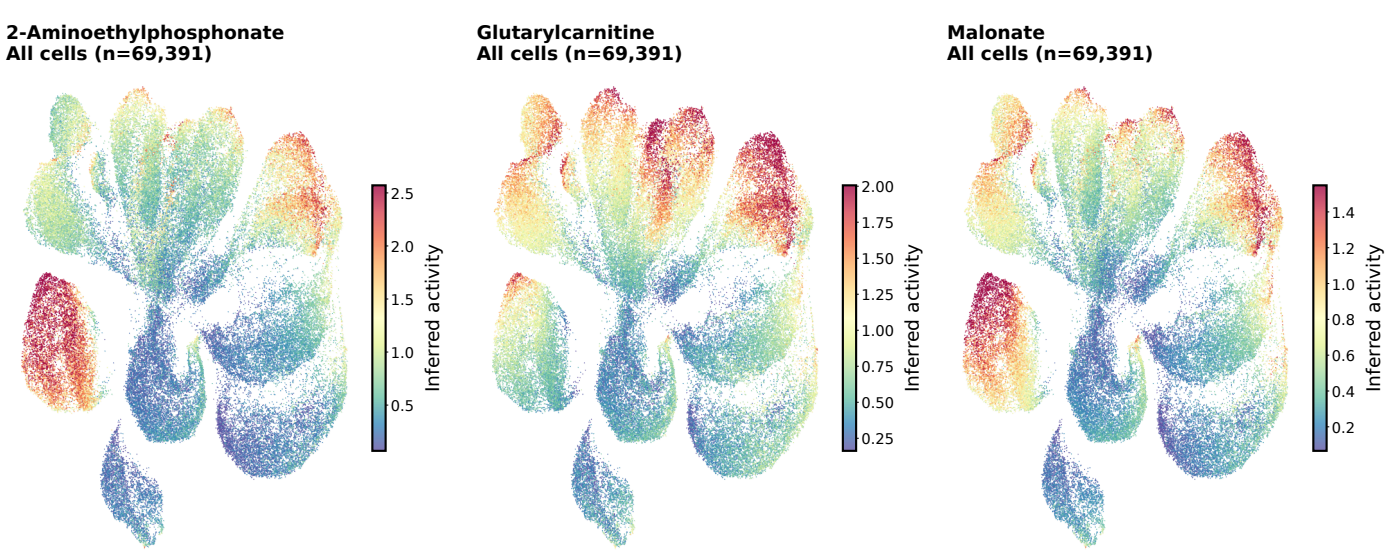
SC GPR-GPA versus scKnife pathway activity

Metastatic − Non-metastatic



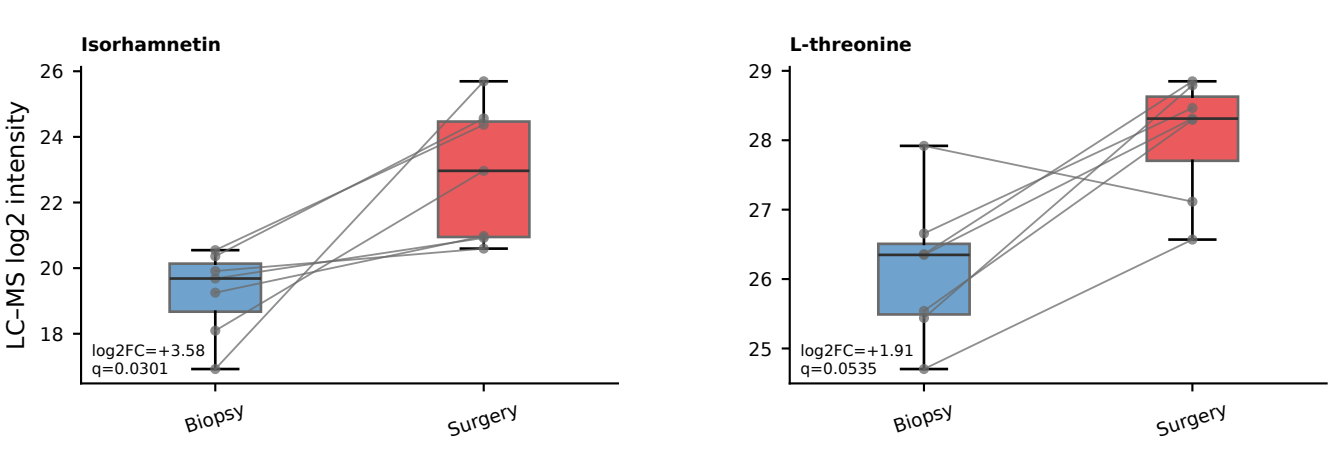
SD Whole-atlas scKNIFE activity

Grey atlas underlay; all cells colored by activity using Spectral_r (independent 2nd–98th percentile scales).



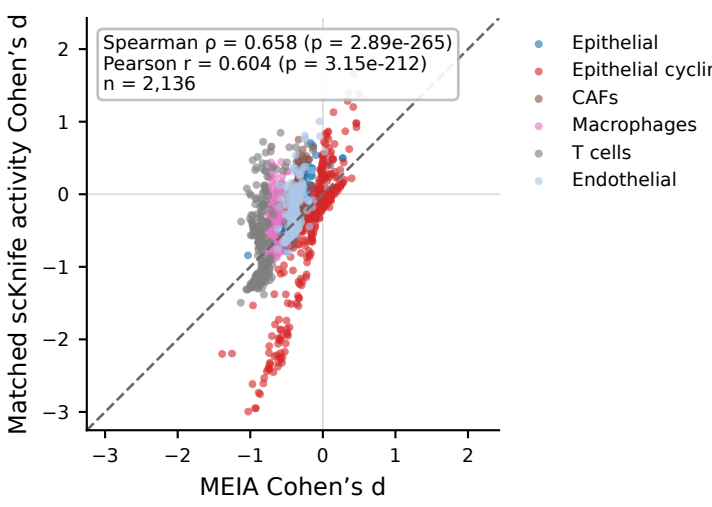
SE Top paired biopsy-versus-surgery LC–MS metabolites

Seven paired patients; corrected named candidates (Isorhamnetin FDR-supported; L-threonine exploratory).



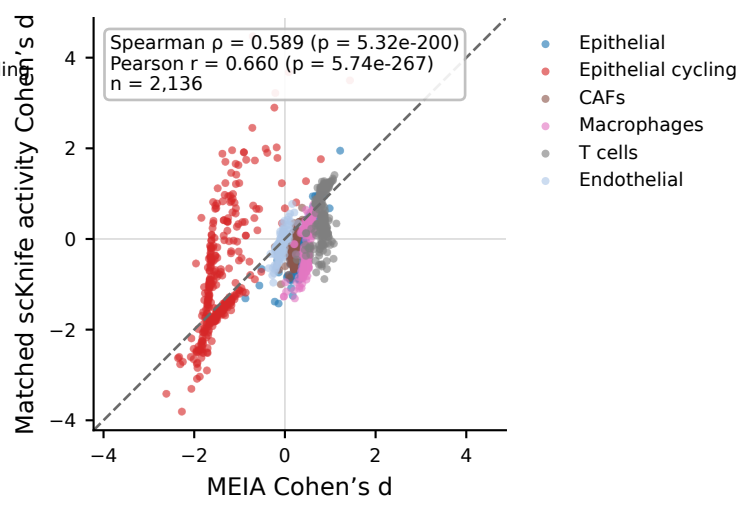
SF MEIA versus matched scKnife activity

Surgery − Biopsy



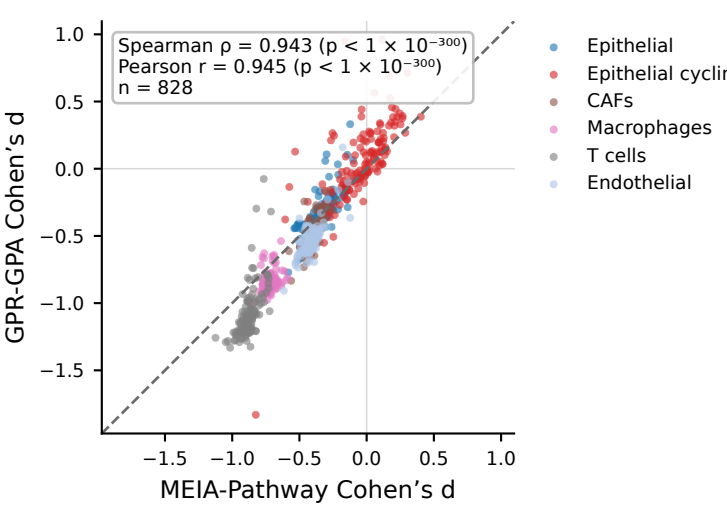
SG MEIA versus matched scKnife activity

Metastatic − Non-metastatic



SH MEIA-Pathway versus GPR-GPA

Surgery − Biopsy



SI MEIA-Pathway versus GPR-GPA

Metastatic − Non-metastatic

